

Article

Sustainable Evaluation Framework for Urban Creative Space: Exploring a Better Way for Urban Development

Shude Song¹, Qiyong Yang²  and Taotao Zou^{3,*}¹ College of Fashion and Design, Donghua University, Shanghai 200051, China; dashu@dhru.edu.cn² Glorious Sun School of Business and Management, Donghua University, Shanghai 200051, China; qiyongyang@mail.dhu.edu.cn³ School of Art and Design, Shanghai Polytechnic University, Shanghai 201209, China

* Correspondence: ttzou@sspu.edu.cn

Abstract

Amid the accelerating waves of global digitalization and the deepening interplay of cultural diversity, urban creative spaces have become pivotal arenas for the digital creative industry—yet a systematic, cross-culturally robust tool for assessing their sustainability remains conspicuously absent. Here, we address this gap by constructing a multi-dimensional evaluation framework derived from a systematic literature review, comprising five primary dimensions—AIGC technology integration, cultural heritage preservation, the economic benefits of the digital cultural industry, ecological synergy and social inclusiveness, and governance and policy support—along with 20 secondary indicators. To enhance methodological rigor, we integrate the Intuitionistic Fuzzy Analytic Hierarchy Process (IFAHF) to determine indicator weights while mitigating the subjective biases inherent in traditional approaches and employ the TOPSIS method to quantitatively assess and rank the creative spaces of five representative cities: London, Shanghai, Los Angeles, Tokyo, and Berlin. Our findings reveal that London leads in comprehensive sustainability, followed closely by Shanghai, with sensitivity analysis confirming the high robustness of the rankings. The originality of this work lies in reconceptualizing AIGC not as a conventional digital instrument but as a core transformative driver embedded within the evaluation architecture, while the application of IFAHF substantially enhances the scientific validity and methodological reliability of the assessment. This research provides an operational diagnostic tool and actionable optimization pathways for advancing the sustainability of urban creative spaces worldwide, offering practical implications for fostering cultural innovation, bridging the digital divide, promoting social inclusiveness, and informing evidence-based urban governance policies.

Keywords: urban creative space; sustainable development evaluation; intuitionistic fuzzy AHP (IFAHF); TOPSIS; multi-dimensional indicator system; international case study



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1. Introduction

With the advent of the big data era, the trend of technological reform has impacted various aspects such as the economy [1], society [2], design and innovation [3]. As a gathering place for cultural design creativity, urban areas have been particularly greatly influenced in the fields of culture, art creativity, and design in recent years. The digital creative industry, as a new industry that integrates traditional cultural creative industries (CCIs) with digital technology, has become an important driving force for promoting

the transformation and upgrading of urban economies and the protection of cultural inheritance [4].

In the series of upgrades, the urban creative space is the key carrier for the establishment and growth of this industry. Since a series of activities and industries are carried out in urban creative spaces, it serves as a key implementation site for the digital economy. As an important implementation location for the digital economy, it is not only a place for the aggregation of creative enterprises and the exchange of talents, but also a diverse ecological environment for the application of digital technology, the revitalization of cultural heritage, and the transformation of economic value. There have been previous studies focusing on the regional aspects of urban creativity and cultural industries [5].

With the accelerated penetration of AIGC and virtual reality, the development model of urban creative spaces is undergoing a subversive transformation fundamentally different from traditional digitalization [6]. Different from traditional digitalization that only realizes online migration of creative activities, AIGC reshapes the sustainable logic of creative spaces at three levels: breaking the geographical boundary of creative factor agglomeration (economic factor), innovating the intergenerational inheritance mode of cultural heritage (cultural factor), and lowering the threshold of public creative participation (social factor). This transformation also brings multiple challenges to the sustainable development of urban creative spaces: (1) The integration of technologies is not deep enough [7]. The application of emerging tools such as AIGC cannot fully balance creative design and traditional artistic traditions; (2) The balance between cultural inheritance and commercial development also needs to be maintained [8]; (3) Economic effects and ecological protection cannot be balanced [9]. The high energy consumption of the operation of creative spaces has led to widespread resource waste; and (4) The policy support and governance mechanism are still not perfect [10]. There is a lack of intellectual property protection, and insufficient supply of digital services, which restricts long-term development. (see urban creative space in Figure 1).



Figure 1. Urban creative space. (Photographed by the author in Donghua University, Shanghai, China).

Regarding these issues and challenges, an assessment of the construction of urban creative spaces should be conducted on a series of problems. Moreover, considering the current social background of urban sustainable development [11], a complete, scientific and replicable evaluation model should be constructed. The sustainable development of urban creative spaces is not a single-dimensional growth [12]. Rather, it involves balancing and coordinating multiple aspects such as the inclusiveness of new technologies, cultural protection and inheritance, economic effects, ecological synergy, and policy governance.

Only by establishing a comprehensive evaluation framework can the current development shortcomings be accurately identified, and resource distribution be optimized in response to the identified problems. Our research aims to promote the transformation of urban creative spaces from simply scale expansion to quality improvement and thereby provide policy implementation and deployment suggestions for urban sustainable development. In response to the shortcomings of existing research, a new research method and evaluation system for urban sustainable development have been explored. The core contributions are reflected in four aspects:

- (a) Reconstructing AIGC as the core driving force for the sustainable development of urban creative spaces, breaking away from its traditional position as a single-scenario auxiliary tool.
- (b) Constructing a five-dimensional integrated evaluation framework that integrates technology, culture, economy, ecology, and governance, solving the problem of the existing research's single dimensionality/ecological–social disconnection.
- (c) Integrating the IFAHP-TOPSIS hybrid method and adding sensitivity analysis to reduce expert subjective bias and enhance the robustness of the evaluation results.
- (d) Designing unified evaluation indicators for global cross-city comparisons, compensating for the existing research's regional specificity and lack of cross-cultural measurement standards. We will further explain the contribution logic behind the results.

The rest of this paper is organized as follows. Section 2 sorts out the theoretical basis and research gaps through a literature review. Section 3 constructs the evaluation indicator system, introduces study samples, data sources and the IFAHP-TOPSIS method. Section 4 presents the indicator weights and city sustainability ranking results. Section 5 discusses the performance of sample cities and common optimization directions. Section 6 summarizes the core findings, contributions and limitations of this study.

2. Theoretical Background and Literature Review

The research on urban creative spaces can be mainly divided into five parts: (1) integration of AIGC technology; (2) cultural inheritance and protection; (3) economic benefits brought by digital cultural industries; (4) ecological collaboration and social inclusiveness; and (5) policy support. This content corresponds to some of the issues we pointed out in Section 1. In the following text, we will describe in detail the explanations of current existing research for these fields.

2.1. AIGC Technology Integration

The integration of AI technology has enabled art and design to be presented in a more diverse form. Many studies have focused on the integration of AI technology into art and design. Shimin Pan et al. [13] proposed integrating AI technology into the design of a traditional intangible cultural heritage tea pot, and proposed a corresponding sustainable evaluation scheme, assessing three dimensions: esthetics, functionality, and emotion. The addition of innovative technology can make the design of intangible cultural heritage pottery more interesting, and it plays a crucial role in the protection of cultural heritage. Xiuhong Li et al. [14] collected information from students, teachers, and enterprises in the art design industry through interviews and open-ended responses regarding how to apply AIGC technology to design and how to design art education. Rebecca Heaton's research is particularly distinctive [15], who handed over the disputes and problems of AI in art education to artificial intelligence to answer. However, regarding the integration of AIGC into education and art, many scholars still have different opinions, and the focus of research varies. Some studies focus on the application of AI in early childhood art education [16], while some focus on AI research [17].

To address these challenges, the core scientific question of this study is how to construct a scientific, globally applicable multi-dimensional evaluation framework to measure the sustainable development level of urban creative spaces in the digital age. Regarding the integration of AIGC in art design, we must admit that it is a disruptive technology. Especially when compared with the era when art could only be created by humans, AI technology provides new inspirations for design and creativity, promoting the birth of new inspirations and innovations [18], which is beneficial to human art design and urban innovation capabilities.

2.2. Cultural Inheritance and Protection

During the process of art design, great importance should be attached to the protection of cultural heritage. The integration of digital technology can enable more cultural heritage to be protected. Zoran Ognjanović et al. [19] suggest adopting digital document management by building an information-based document management system, and, taking Serbia as an example, proved the significant role of digital technology in the protection of cultural heritage. In addition to protecting texts such as documents through digitalization, there are also related studies on photography technology and color capture technology, which enable the colors and images of cultural relics to be greatly restored. Przemyslaw Korytkowski and Agnieszka Olejnik-Krugly [20] conducted shooting experiments on six different RGB color files for the cultural heritage protection of museums, thereby verifying that for different cultural relics, the corresponding most suitable RGB color files need to be used for restoration. Injung Song et al. [21] restored a traditional color of an intangible cultural heritage in Korea (presented on ancient buildings); six domain experts tried different types of color mixing and calculated color accuracy and found that the color tone played a crucial role in the color mixing process.

Figure 2 illustrates a digital approach to the protection of intangible cultural heritage in the field of clothing as a case study. The School of Fashion and Design, Donghua University, has digitized the design archives of the ethnic costumes of the minority groups in Liupanshui City, Guizhou Province, China. Based on the local culture and clothing styles, students from Donghua University also have designed brand-new and distinctive clothing.

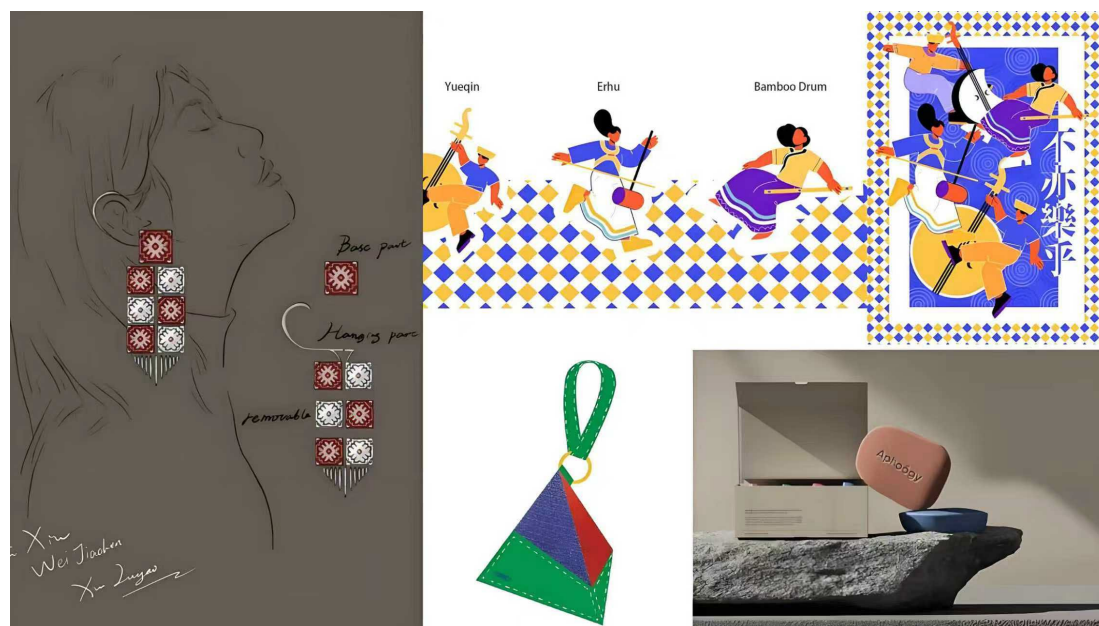


Figure 2. Ethnic minority clothing design.

2.3. The Economic Benefits of the Digital Cultural Industry

The changes in technology have not only brought numerous changes to our lives but also brought about a completely new presentation of artistic forms and results and further enhanced the benefits of the digital economy. The digitization of creativity and art design, in the context of the current digital age, can bring more job opportunities to society, which is a potential benefit. At the same time, society can create more commercial income through artistic activities and the market. Some artistic activities have great commercial potential and value and can effectively provide economic benefits.

Xu Chen et al. [22] mentioned in their article that digital technology has broken the barriers between online and offline activities, and the resulting agglomeration effect can promote the increase in the output value of the creative industry and bring certain employment contributions. At the same time, the constructed digital platform can also improve the efficiency of income acquisition. Sergio Barile et al., in their study [18], proposed that enterprises, the public, and the government can cooperate to build an ecosystem and create economic effects, such as creating some creative products and commercializing them. Jing Huang et al. [23] emphasized in her research that market demand and policy guidance are two main factors that can guide the growth of the industrial economy, which is a new inspiration for the digital economy industry.

2.4. Ecological Synergy and Social Inclusiveness

Ecological synergy and social inclusiveness are the core connotations of the sustainable development of urban creative spaces, which are deeply rooted in the classic theories of circular economy and urban resilience: the circular economy theory requires creative spaces to minimize the negative impact on the ecological environment through material recycling and full life cycle low-carbon operation, while the urban resilience theory emphasizes that creative spaces should enhance the inclusiveness of urban development by reducing the digital divide and improving multi-stakeholder participation, forming a harmonious symbiosis between the space, the environment and the society.

Haiyong Li et al. [24] proposed in their research that in the process of art design, designers and artists should maintain a sense of environmental protection responsibility and maintain a balance between artistic innovation and ecological responsibility. Salim Barbhuiya et al. [25] considered the life cycle of building materials, constructed a life cycle assessment model, and evaluated different building materials, which effectively calculates the usable period of a building and can also be applied to cultural buildings. The design of creative buildings should pay more attention to their service life, and buildings with a longer service life can reduce the waste of building materials. Roman Meinhold et al. [26] pointed out in their research that artificial intelligence (AI), blockchain, and electric vehicles (EVs) in digital technology bring a new impetus to environmental protection, which is also applicable to the art design and creative fields (creative design space needs to be well integrated with technology). In addition, in the field of architecture, the addition of biomimicry can also effectively reduce waste. Through 3D printing methods, sustainable buildings can be realized, which is also indispensable in the construction of creative spaces [27]. Therefore, in the challenges of building urban creative spaces, it is very important to reduce the impact of buildings on the environment from all aspects [28].

At the same time, the development of urban creative spaces requires the support of social inclusiveness. In the process of urban digital transformation, it is necessary to fully consider the accessibility, availability, and responsiveness of services, and consider some vulnerable groups (such as the elderly) when facing digital technology—can they adapt? Or can new technologies meet the requirements of as many groups as possible [29]. In the field of art education, the introduction of virtual reality (VR) technology can cultivate

sustainable responsibility and digital leadership. Yuxin Zhang et al. [30] proposed the TAM-SDT-DL framework, which improves the quality of education by reducing inclusivity differences and digital skills gaps, providing a reference model for policy implementers. Most of these studies, however, prioritize total economic output over ecosystem-level value co-creation and industrial spillover effects. They also decouple economic benefits from social benefits and lack a unified evaluation framework that adapts to different economic contexts of global cities.

2.5. Policy Support

The development of digital creative spaces also requires the support of the government. Tamsyn Dent et al. [31] conducted an in-depth study of a case in the Netherlands and found that although the local area was able to attract talents to start businesses, it was unable to retain the talents. Moreover, the content funded by the government was significantly different from the construction of the urban creative ecosystem, and it could not accurately meet the needs of people. Fanyu Zhang and Helen Kennedy [32] explored the issues at the enterprise level: does the government need to provide certain assistance to a cultural and creative enterprise during its operation? In their research, the authors called for policy support and authorized the enterprise to expand its social and cultural influence. In the field of fashion creativity, project rewards can provide economic support for the innovative activities of enterprises [33] and can also amplify the cluster effect and provide policy and economic support for the development of the creative industry.

What the government brings not only includes economic support but also can similarly enhance the public's support for the creative industry and is conducive to sustainable economic development. The government's regulation of the industry plays a very important promoting role.

There is still much research to be done on urban creative industries. The regional distribution of creative industries has specific social regional characteristics [34]. Big data and artificial intelligence technologies provide support for supply chain innovation and international marketing strategies for small- and medium-sized cultural and creative enterprises [35]. Local governments can also enhance the sustainable development of creative design industries [36]. These studies, however, focus heavily on financial subsidy policies while ignoring institutional guarantees like intellectual property protection and cross-departmental governance. They also lack cross-institutional comparative analysis and research on policy synergy across technological, cultural, and ecological dimensions.

Meanwhile, recent studies have also confirmed the applicability and optimization potential of the TOPSIS method in sustainable development evaluation. Du et al. [37] proposed a multi-scale urban sustainability evaluation method, pointing out that the TEA-IS method can effectively solve the problem of indicator heterogeneity in cross-regional evaluation. Additionally, Jarosław Wątróbski et al. focused on urban community sustainability, using DARIA-TOPSIS and applying it to assess the community spatial sustainability rankings of 26 cities, and conducting time series analysis [38]. These related existing studies provide important methodological references for this research to integrate the IFAHP-TOPSIS method and conduct global cross-city creative space sustainability evaluation.

Existing studies have built a solid foundation for understanding urban creative space development by exploring key dimensions including AIGC application, cultural heritage protection, digital economic benefits, ecological sustainability, and policy support. As shown in Table 1, these works have laid out technical paths for digital cultural preservation, verified the economic agglomeration effects of digital technologies, and highlighted the role of policy incentives in nurturing creative ecosystems. Recent methodological research has also confirmed the effectiveness of the IFAHP-TOPSIS hybrid model in handling fuzzy

multi-attribute decision-making in sustainable development evaluation [39–41], providing a reliable analytical tool for this study [42].

Table 1. Core differences between this study and the existing literature.

Dimension	Existing Literature Characteristics	Innovations
AIGC Positioning	Treated as an incremental auxiliary tool for single scenarios	Framed as a core transformative driver reshaping creative space logic
Evaluation Framework	Single-dimensional or partially integrated; ecological–social decoupling	Integrated 5 core dimensions
Methodological Rigor	Rarely uses IFAHP to reduce expert subjectivity; no sensitivity analysis	Hybrid IFAHP-TOPSIS with sensitivity analysis for result robustness
Applicability	Region-specific; lack of cross-cultural measurement standards	Cross-city comparative design with unified indicators for global adaptability

Despite these advances, a framework for urban creative space sustainability remains absent in the current scholarship. Most existing frameworks either marginalize AIGC’s transformative role or fail to integrate ecological synergy with social inclusiveness, creating disconnects between theoretical dimensions and practical evaluation needs. Furthermore, the lack of unified cross-cultural measurement standards and insufficient attention to methodological robustness limit the generalizability of existing findings.

This study addresses these gaps by constructing an integrated five-dimensional evaluation system, embedding AIGC as a core driving force, and applying a rigorous hybrid method to conduct cross-city comparative analysis—efforts that aim to bridge the divide between fragmented existing research and the need for systematic, actionable evaluation tools for global urban creative space development.

3. Materials and Methods

3.1. Sustainable Evaluation Indicator System

To scientifically assess the sustainable development level of urban creative spaces, we construct a multi-dimensional evaluation system based on the literature review, which covers five core dimensions: (1) AIGC technology integration, (2) cultural inheritance and protection, (3) economic benefits of digital cultural industries, (4) ecological synergy and social inclusiveness, and (5) governance and policy support, based on the findings of Section 2.

The construction of the evaluation system adheres to the principles shown in Figure 3. These principles are reflected in the close alignment of each indicator with theoretical foundations and empirical results of existing studies. Systematic evaluation model is embodied in the comprehensive coverage of technical driving forces, cultural core, economic support, ecological bottom line, and institutional guarantee of sustainable development, while its operability is ensured by the clear definition of each indicator and accessible data sources.

In addition, sustainability runs through the entire system, focusing on long-term evolution, intergenerational inheritance, and coordinated development of the economy, society, and environment. This system aims to comprehensively reflect the development status of urban creative spaces and provide a reliable tool for identifying development bottlenecks and optimizing improvement paths.

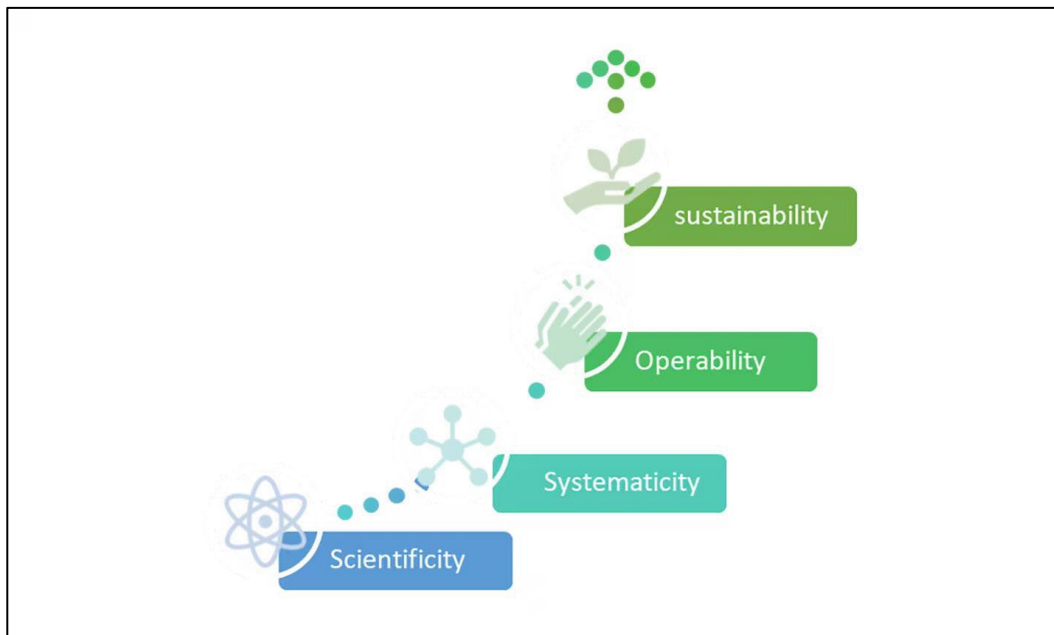


Figure 3. Principles of evaluation model. (see source in Supplementary).

1. AIGC Technology Integration

As a disruptive driver for the transformation of urban creative spaces, AIGC integration is a key prerequisite for their sustainable innovation. This dimension focuses on the practical application effect and long-term adaptability of AIGC technology. AIGC technology penetration depth reflects the coverage and application depth of digital technologies such as AIGC and big data in creative scenarios like creative generation and immersive experiences, an important manifestation of technology-driven sustainable innovation [1,2,6]. Virtual agglomeration synergy efficiency enhances the sustainable agglomeration of creative factors by breaking geographical boundaries [6,11,22], while technology iteration resilience helps avoid path dependence amid technological updates and external shocks [1,18].

2. Cultural Inheritance and Protection

Cultural inheritance and protection are the core connotations of urban creative spaces' sustainability, key to maintaining their uniqueness and vitality. This dimension focuses on traditional culture's digital protection, public participation, and intergenerational inheritance, with five secondary indicators covering the whole process. The digital coverage rate of traditional culture realizes the permanent preservation and wide dissemination of cultural heritage through digital means like archiving and VR experiences [4,5,19], while the public participation rate in inheritance strengthens the social identity of cultural heritage [4,6,30]; the integrity of digital cultural resources ensures standardized management and reuse [5,19]. The integration rate of traditional elements is one of the most important parts, which balances inheritance and innovation in creative works [3,21], and intergenerational continuity guarantees cultural vitality across generations [3,4,13].

3. Economic Benefits of Digital Cultural Industry

The economic benefits of digital cultural industries provide an important material guarantee for urban creative spaces' sustainability, reflecting their value conversion and regional economic driving capacity, and the dimension sets three secondary indicators around core economic effects. The creative industry's GDP contribution rate directly reflects economic sustainability and its role in urban economic transformation [2,6]. The economic output of value co-creation embodies sustainable benefits from collaborative development of the innovation ecosystem and the employment-driving effect highlights both economic

value and social responsibility by boosting direct and indirect employment [2,3,22]. All indicators comprehensively measure economic sustainability from output contribution, collaborative benefits and social effects.

4. Ecological Synergy and Social Inclusiveness

Ecological synergy and social inclusiveness reflect the harmonious coexistence of creative spaces with the environment and society, covering ecological and social sustainability. Among all the indicators, the multi-subject collaboration frequency shows the symbiotic development of the innovation ecosystem [1,6,18], and the digital skill training coverage rate reduces the digital divide and enhances social inclusiveness [2,4,29]. The green energy utilization rate [4] and low-carbon operation level [6] show how the cities reduce their environmental impact via renewable energy application. The public satisfaction degree reflects adaptability to public needs [31].

5. Governance and Policy Support

Governance and policy support provide institutional guarantees for the standardized operation and long-term development of urban creative spaces, which focuses on policy support and governance efficiency. Intellectual property protection degree safeguards innovation enthusiasm through copyright registration, infringement handling, and policy implementation [2,5,32], while policy support intensity offers economic support and guidance via subsidies, tax incentives, and special funds [33]. The digital service level and cross-subject collaborative governance separately improve operational efficiency through government digital services, data sharing [5,19,22] and joint standard-setting, supervision, and resource allocation [6,36].

The above evaluation system comprehensively covers the key factors affecting the sustainable development of urban creative spaces, forming a multi-dimensional, multi-level, and systematic evaluation framework (Table 2 below). Each indicator includes definitions, a data description, and literature support, which can effectively measure the sustainable development level of urban creative spaces.

Table 2. Sustainable evaluation model for urban creative spaces.

First-Level Indicators	Second-Level Indicators	Indicator Description	Measurement Method	Label
AIGC Technology Integration (G_1)	AIGC Technology Penetration Depth	Extent of AIGC adoption across the creative value chain (e.g., ideation, production, visualization) in urban creative spaces.	Derived from industry whitepapers and enterprise annual reports; measured by the percentage of creative enterprises using AIGC tools and the share of AI-enabled creative scenarios.	x_1
	Virtual Agglomeration Synergy Efficiency	Efficiency of cross-regional resource sharing and collaborative innovation among creative entities via digital platforms.	Compiled from regional cultural industry association reports; measured by the number of annual cross-regional collaborative projects and the conversion rate of shared resources into tangible outputs.	x_2
	Technology Iteration Resilience	Ability of creative spaces to adapt to AIGC technological upgrades and avoid path dependence.	Extracted from technology innovation monitoring reports; measured by the frequency of core technology updates and the proportion of independently developed AIGC applications.	x_3

Table 2. Cont.

First-Level Indicators	Second-Level Indicators	Indicator Description	Measurement Method	Label
Cultural Inheritance and Protection (G_2)	Digital Coverage Rate of Traditional Culture	Proportion of intangible cultural heritage (ICH) and historical resources digitized for archival and dissemination.	From municipal cultural and tourism bureau statistical bulletins; measured by the percentage of ICH items with complete digital archives.	x_4
	Public Participation Intensity in Cultural Inheritance	Level of public engagement in digital cultural heritage activities (e.g., virtual exhibitions, digital creation).	From closed-source database of this Shanghai Philosophy and Social Sciences Planning Project; measured by the annual number of public visits to digital cultural platforms and participation in online cultural creation events.	x_5
	Integrity of Digital Cultural Resources	Standardization and reusability of digitized cultural resource archives across platforms.	Assessed by expert panel based on cultural resource standardization reports; measured by the percentage of archives complying with national digital resource standards.	x_6
	Integration Rate of Traditional Elements	Degree of organic integration of traditional cultural symbols and craftsmanship into modern creative products.	Sampled from creative industry product catalogs; measured by the percentage of products incorporating traditional cultural elements.	x_7
	Intergenerational Continuity of Cultural Inheritance	Acceptance and inheritance willingness of young people (18–35 years) towards digital cultural resources.	From youth cultural consumption reports published by municipal statistics bureaus; measured by the penetration rate of digital cultural products among the youth population.	x_8
Economic Benefits of Digital Cultural Industry (G_3)	Creative Industry GDP Contribution Rate	Share of the digital creative industry's added value in the city's total GDP.	Extracted from municipal statistical yearbooks (2023–2024).	x_9
	Economic Output of Value Co-Creation	Total economic benefits generated by collaborative projects among government, enterprises, and institutions.	Compiled from regional cultural industry economic analysis reports; measured by the total output value of collaborative innovation projects.	x_{10}
	Employment-Driving Effect	Capacity of the digital creative industry to drive direct and indirect employment.	From municipal human resources and social security bureau employment reports; measured by the number of direct jobs and the growth rate of indirect jobs in the creative industry chain.	x_{11}
Ecological Synergy and Social Inclusiveness (G_4)	Multi-Subject Collaboration Frequency	Frequency of collaborative governance and innovation activities among government, enterprises, and universities.	From regional innovation ecosystem development reports; measured by the annual number of multi-stakeholder collaborative projects.	x_{12}
	Digital Skill Popularization Level	Extent of digital creative skill promotion among vulnerable groups (e.g., the elderly, low-income groups).	From municipal digital inclusion action reports; measured by the number of digital skill training sessions held and the number of participants.	x_{13}
	Green Energy Utilization Rate	Proportion of renewable energy used in the operation of key creative spaces.	From municipal ecological environment bureau carbon emission reports; measured by the percentage of renewable energy consumption in total energy use.	x_{14}
	Public Satisfaction with Creative Spaces	Comprehensive public evaluation of the service quality and accessibility of urban creative spaces.	From closed-source database of this Shanghai Philosophy and Social Sciences Planning Project; aggregated from public satisfaction surveys conducted by authorized third-party research institutions.	x_{15}
	Low-Carbon Operation Level	Degree of achievement of low-carbon operation goals in creative spaces (e.g., waste recycling, carbon reduction).	From enterprise environmental responsibility reports and green building certification data; measured by the resource recycling rate and the completion rate of carbon emission reduction targets.	x_{16}

Table 2. Cont.

First-Level Indicators	Second-Level Indicators	Indicator Description	Measurement Method	Label
Governance and Policy Support (G ₅)	Intellectual Property Protection Degree	Effectiveness of intellectual property (IP) protection for creative works (e.g., copyright registration, infringement handling).	From municipal IP office annual reports; measured by the copyright registration rate of creative works and the average processing time of IP infringement cases.	x_{17}
	Policy Support Intensity	Financial and institutional support provided by the government to the digital creative industry.	Compiled from municipal cultural industry policy documents and fiscal expenditure reports; measured by the amount of special funds for the creative industry and the coverage rate of preferential policies.	x_{18}
	Digital Public Service Level	Sophistication of digital public services (e.g., data sharing, technical consulting) provided to creative enterprises.	From municipal government digital service evaluation reports; measured by the coverage rate of digital public services and the satisfaction score of enterprises.	x_{19}
	Cross-Subject Collaborative Governance Efficiency	Efficiency of collaborative governance among government departments, industry associations, and enterprises.	From municipal governance modernization reports; measured by the completion cycle of industry standard formulation and the response efficiency of collaborative supervision.	x_{20}

Due to the ambiguity and uncertainty of subjective judgments in the evaluation process, and the differences in the importance of each indicator, it is necessary to adopt scientific evaluation methods to determine the weight of each indicator and process the evaluation information. Therefore, in Section 4, the Intuitive Fuzzy AHP (IFAHP) method will be used to scientifically determine the weight of each indicator and will be combined with the TOPSIS to handle the fuzziness and randomness in the evaluation process, to realize the accurate and objective evaluation of the sustainable development level of urban creative spaces.

3.2. Method: Intuitive Fuzzy AHP and TOPSIS

To address the fuzziness and uncertainty of subjective judgments in the sustainable evaluation of urban creative spaces, we combine the Intuitive Fuzzy Analytic Hierarchy Process (IFAHP) and Technique for Order Preference by Similarity to Ideal Solution (TOPSIS).

The IFAHP effectively handles ambiguous information in expert evaluations compared with the traditional fuzzy AHP [43], by introducing intuitionistic fuzzy numbers (IFNs) that simultaneously consider membership degree, non-membership degree, and hesitation degree, making the weight determination more in line with the actual decision-making process. However, TOPSIS, as a classic multi-attribute decision-making method [39,40,44], sorts alternatives by calculating the proximity between each evaluation object and the positive or negative ideal solutions, which is suitable for the comprehensive ranking of multi-indicator evaluation systems. The combination of the two methods (Figure 4) integrates the advantages of handling fuzzy uncertainty and multi-attribute sorting, ensuring that the principles mentioned in Figure 4 are reflected in the evaluation results.

Based on the constructed sustainable evaluation system, a three-level hierarchical structure is established (Figure 5), covering a target layer (sustainable evaluation of urban creative space), goal layer (five first-level indicators) and indicator layer (twenty secondary indicators), which provides a clear framework for subsequent weight calculation and evaluation implementation.

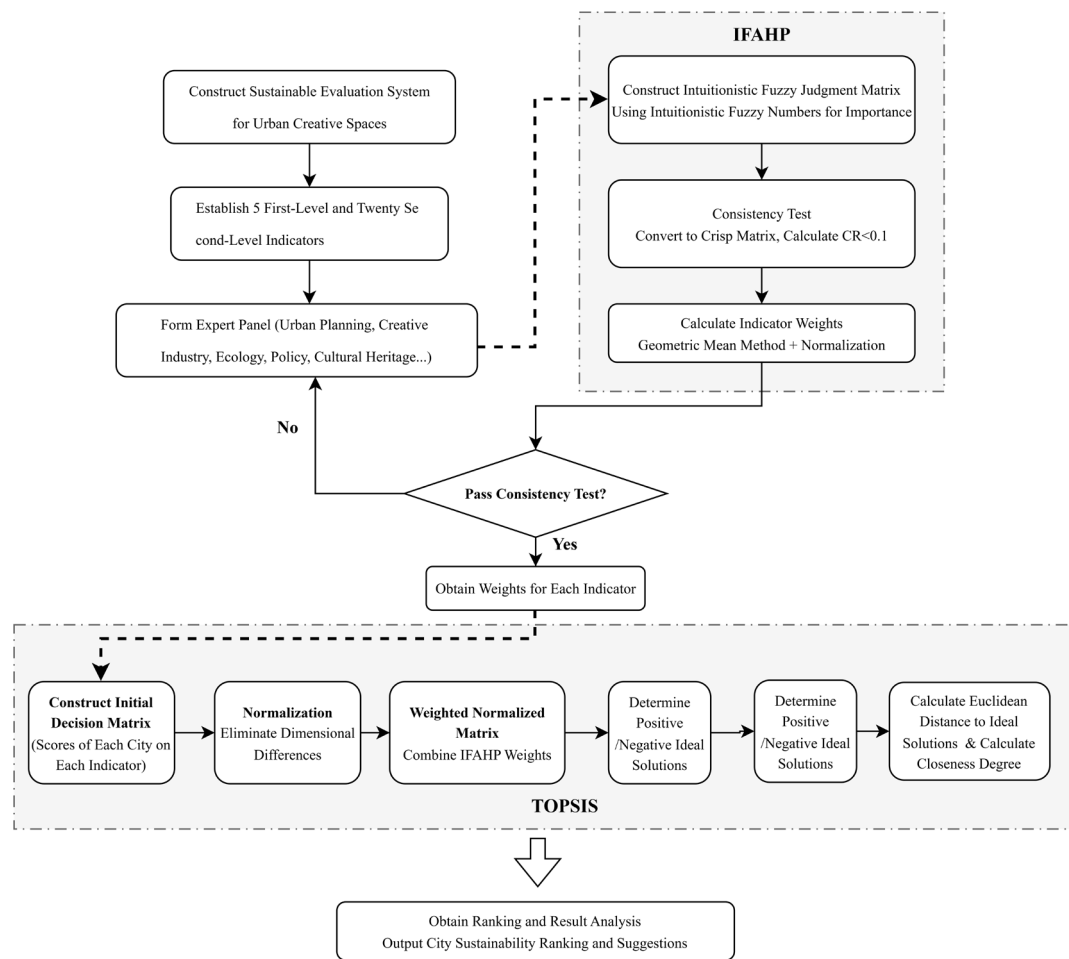


Figure 4. Process of IFAHP and TOPSIS evaluation.

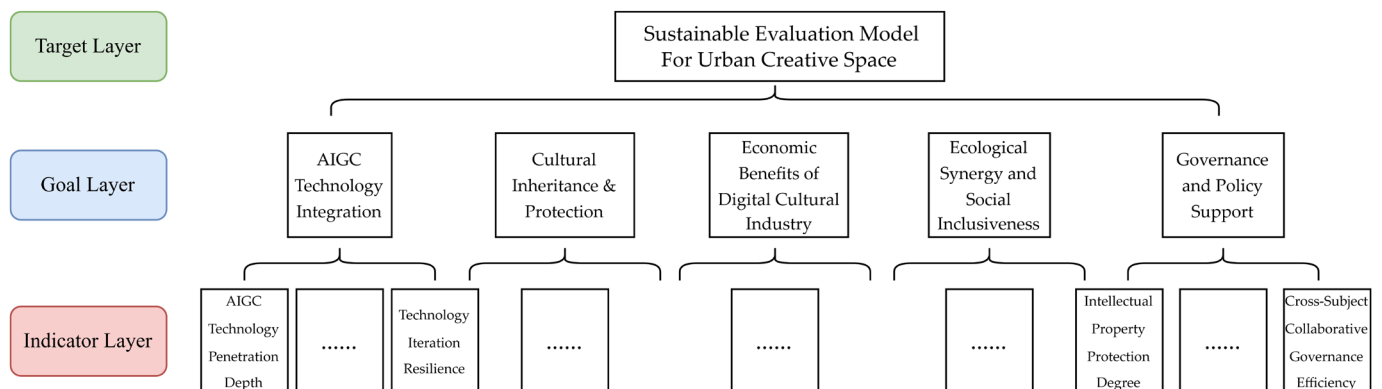


Figure 5. Display image of the evaluation model.

We invited 15 experts from urban planning, the digital creative industry, ecological sustainability and other fields to conduct pairwise comparisons of indicators at the same level. The importance degree was expressed by intuitionistic fuzzy numbers $a_{ij} = (\mu_{ij}, \nu_{ij})$, where μ_{ij} is the membership degree (support degree of indicator i being more important than indicator j), ν_{ij} is the non-membership degree (opposition degree), and the hesitation degree $\pi_{ij} = 1 - \mu_{ij} - \nu_{ij}$ (satisfying $0 \leq \mu_{ij} + \nu_{ij} \leq 1$). The constructed intuitionistic fuzzy

judgment matrix $A = (a_{ij})_{n \times n}$ (where n is the number of indicators at the current level), and satisfies the reciprocal property as follows:

$$\begin{cases} a_{ij} = (\mu_{ij}, \nu_{ij}) \\ a_{ji} = (\mu_{ji}, \nu_{ji}) \\ a_{ii} = (0.5, 0.5) \end{cases} \quad (1)$$

Experts' linguistic evaluations were converted into intuitionistic fuzzy numbers based on the semantic correspondence table (Table 3) to ensure the standardization of evaluation data.

Table 3. Semantic correspondence table for intuitionistic fuzzy evaluation.

Linguistic Evaluation	$(\mu_{ij}, \nu_{ij}, \pi_{ij})$
Extremely Important (EI)	(0.90, 0.05, 0.05)
Very Important (VI)	(0.80, 0.10, 0.10)
Fairly Important (FI)	(0.70, 0.20, 0.10)
More Important (MI)	(0.60, 0.25, 0.15)
Equally Important (EI)	(0.50, 0.30, 0.20)
Less Important (LI)	(0.40, 0.45, 0.15)
Very Less Important (VLI)	(0.30, 0.60, 0.10)
Extremely Less Important (ELI)	(0.10, 0.85, 0.05)

To ensure the logical rationality of the judgment matrix, a consistency test was conducted. First, the intuitionistic fuzzy matrix was converted into a crisp matrix using the score function $S(a_{ij}) = \mu_{ij} - \nu_{ij}$; then, the consistency index CI and random consistency ratio CR were calculated:

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (2)$$

$$CR = \frac{CI}{RI} \quad (3)$$

where λ_{max} is the maximum eigenvalue of the crisp matrix, and RI is the random consistency index. When $CR < 0.1$, the matrix meets the consistency requirement; otherwise, it needs to be adjusted by experts.

After passing the consistency test, the geometric mean method was used to calculate the weight of each indicator. The intuitionistic fuzzy weight obtained by normalization was converted into a crisp weight through the score function, and the final normalized indicator weight $W = [w_1, w_2, \dots, w_n]$ (satisfying $\sum_{i=1}^n w_i = 1$) was obtained. The core conversion formula of the crisp weight is calculated as follows:

$$w_i^* = \frac{S(w_i) + 1}{2} \quad (4)$$

After obtaining the weights, the TOPSIS was used to rank the sustainable development level of urban creative spaces based on the weights obtained from the IF-AHP. Assume there are m sample cities and n evaluation indicators, and x_{ij} denotes the evaluation value of the i -th city on the j -th indicator. First, the initial decision matrix was constructed.

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \quad (5)$$

To eliminate the influence of indicator dimensions, the range normalization method was adopted. For positive indicators (higher values indicate better performance), the values were calculated as follows:

$$z_{ij} = \frac{x_{ij} - \min_{i=1}^m x_{ij}}{\max_{i=1}^m x_{ij} - \min_{i=1}^m x_{ij}} \quad (6)$$

Combining the indicator weights W obtained from IFAHP, the weighted normalized matrix $V = (v_{ij})_{m \times n}$ was constructed:

$$v_{ij} = z_{ij} \times w_j \quad (7)$$

On this basis, the positive ideal solution V^+ (maximum value of each indicator in the weighted normalized matrix) and negative ideal solution V^- (minimum value of each indicator in the weighted normalized matrix) were determined:

$$V^+ = [v_1^+, v_2^+, \dots, v_n^+] = [\max_{i=1}^m v_{i1}, \max_{i=1}^m v_{i2}, \dots, \max_{i=1}^m v_{in}] \quad (8)$$

$$V^- = [v_1^-, v_2^-, \dots, v_n^-] = [\min_{i=1}^m v_{i1}, \min_{i=1}^m v_{i2}, \dots, \min_{i=1}^m v_{in}] \quad (9)$$

The Euclidean distances from each city to the positive ideal solution (D_i^+) and negative ideal solution (D_i^-) were calculated, respectively, using the following formula:

$$D_i^\pm = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^\pm)^2} \quad (10)$$

Finally, the closeness degree C_i of each city to the ideal solution was calculated by the formula $C_i = \frac{D_i^-}{D_i^+ + D_i^-}$ and the cities were ranked in descending order of C_i to obtain the sustainable development level ranking.

3.3. City Selection and Data Collection

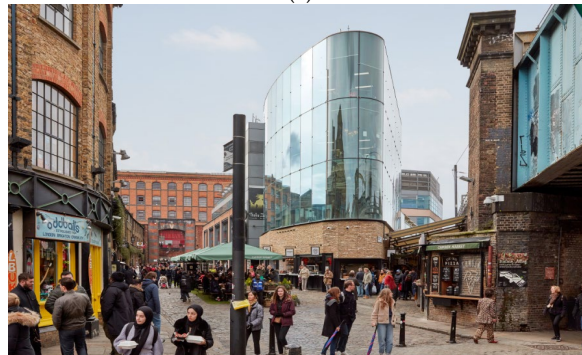
To verify the constructed sustainable evaluation system for urban creative spaces, we selected 5 representative international cities with mature creative spaces as research samples. The selection of sample cities adheres to the principles of representativeness, data accessibility, and diversity of development models, referring to the evaluation standards of the UNESCO Creative Cities Network [6] and existing research on global creative industry agglomeration [7,45]. Five cities were finally selected: Shanghai (China), London (UK), Los Angeles (USA), Tokyo (Japan), and Berlin (Germany). (Figure 6a–e shows examples of the creative space of the five cities).

- Shanghai: As a leading creative city in East Asia, it has formed mature creative clusters, integrating digital creative industries, cultural heritage protection, and policy support [8,9]. Its unique advantage lies in the integration of traditional cultural resources with digital technology, making it a typical case of a “cultural inheritance plus technological innovation” city.
- London: A global creative industry benchmark, featuring diverse creative spaces from art districts [46] to street art performance [47], with strong market-driven economic benefits and social inclusiveness.
- Los Angeles: The global center of film and television creative industries, anchored by Hollywood [48]. Its creative spaces are dominated by film production bases, digital special effects studios, and cultural entertainment clusters driven by the film industry and deep integration of AIGC in content creation.
- Tokyo: The creative industry spaces in the Tokyo area are shaped by multiple factors, including both internal and external elements such as local culture and economic conditions, as well as the influence of the real estate market [49]. The environmental economy also has important considerations for Tokyo’s creative industries [50].

- Berlin: Generally speaking, cities in Europe have their own unique cultures [51]. Known for its inclusive creative environment and policy support for start-ups, its creative spaces emphasize social inclusiveness and cross-border collaborative governance [52,53].



(a)



(b)



(c)



(d)

Figure 6. Cont.



Figure 6. (a). Example of Shanghai creative space. (b) Example of London creative space. (c) Example of Los Angeles creative space. (d) Example of Tokyo creative space. (e) Example of Berlin creative space. (see source in Supplementary).

To ensure the objectivity of weight determination, 15 experts are invited to participate in the pairwise comparison of indicators. The team covers five core fields related to the evaluation system, with a balanced gender ratio and diverse professional backgrounds (see Table 4).

Table 4. Composition of the expert evaluation team.

Gender	Research Field	Occupation	Number of Experts
Male	Urban Planning	University Professor	3
	Digital Creative Industry	Enterprise Manager	2
	Ecological Sustainability	Institute Researcher	2
Female	Policy Research	Government Department Specialist	3
	Cultural Heritage Protection	Museum Curator	2
	Comprehensive Evaluation	Independent Consultant	3
Total			15

The evaluation is determined based on the data obtained through multiple channels to ensure accuracy.

(a) Municipal Government Statistical Reports ($x_4, x_8, x_9, x_{11}, x_{14}, x_{17}, x_{18}, x_{19}, x_{20}$)

These reports provide the foundational objective data for the evaluation system, eliminating subjective bias. Data are extracted from the statistical yearbooks of the five sample cities, as well as official bulletins from municipal bureaus of culture and tourism, statistics, human resources, ecological environment, and intellectual property.

For cross-country consistency, data from non-Chinese cities are sourced from their national statistical offices and cultural policy departments, with currency and unit conversions standardized using OECD exchange rates and statistical guidelines.

(b) Industry Association Reports ($x_1, x_2, x_3, x_7, x_{10}, x_{12}, x_{16}$)

These reports reflect real-world operational performance of creative enterprises and industrial clusters, capturing micro-level dynamics omitted in macro statistics. Data are compiled from annual reports of global and regional cultural/creative industry associations (e.g., the UK Creative Industries Council and China's Cultural Industry Association) and whitepapers published by leading creative enterprises in each city. Key metrics (e.g., AIGC adoption rate and collaborative project volume) are cross-checked across three independent industry reports to ensure consistency.

(c) Database from Shanghai Philosophy and Social Sciences Planning Project (x_5, x_{15})

Encompassing exclusive, high-resolution data collected specifically for this research via the supported funding project, this database aggregates anonymized micro-data from two channels: (1) behavioral data from digital cultural platforms; and (2) aggregated

satisfaction scores from third-party market research institutions commissioned by the project. All data are de-identified to protect privacy, and only aggregated statistical results are used in the evaluation, complying with ethical research standards.

(d) Expert Panel Assessment (x_6, x_{13})

This assessment converts non-quantifiable indicators (e.g., resource integrity) into measurable values using domain expertise, ensuring that the evaluation captures intangible dimensions of sustainability. The expert panel, led by scholars mainly from universities and professional organizations, conducts blind assessments of standardized materials (e.g., cultural resource archives and digital skill training programs).

Each indicator is scored on a 0–100 scale, with the final score calculated as the average of expert ratings after excluding outliers (obeying the 3σ principle). The intra-class correlation coefficient of expert scores is 0.89, confirming high inter-rater reliability.

Based on the above information collection process, we have summarized the relevant data and scores of all 20 indicators for each of the 5 cities. The following table presents the corresponding data from 5 cities. The initial evaluation data of 20 secondary indicators for 5 sample cities (Shanghai, London, Los Angeles, Tokyo, and Berlin) are generated based on their actual development characteristics. All indicators are positive indicators, which means that higher values indicate better performance, and the data range is set to 0–100 (integer) by the expert team to ensure logical consistency and discriminability (Table 5 below shows the score).

Table 5. Data of evaluation model in Shanghai, London, Los Angeles, Tokyo, Berlin.

First-Level Indicators	Indicators \ City	Shanghai	London	Los Angeles	Tokyo	Berlin
AIGC Technology Integration G_1	x_1	75	88	82	70	68
	x_2	72	90	85	69	71
	x_3	68	85	78	73	65
Cultural Inheritance and Protection G_2	x_4	85	75	60	70	58
	x_5	80	72	55	65	52
	x_6	78	70	58	68	55
	x_7	82	68	62	63	50
	x_8	88	73	53	60	48
Economic Benefits of Digital Cultural Industry G_3	x_9	70	78	92	65	60
	x_{10}	68	75	90	62	57
	x_{11}	72	73	88	68	63
Ecological Synergy and Social Inclusiveness G_4	x_{12}	70	76	65	80	82
	x_{13}	65	73	60	78	85
	x_{14}	68	70	55	90	75
	x_{15}	75	80	72	85	83
	x_{16}	70	72	58	92	78
Governance and Policy Support G_5	x_{17}	80	85	88	75	70
	x_{18}	85	78	65	72	63
	x_{19}	82	83	70	76	68
	x_{20}	78	80	68	73	71

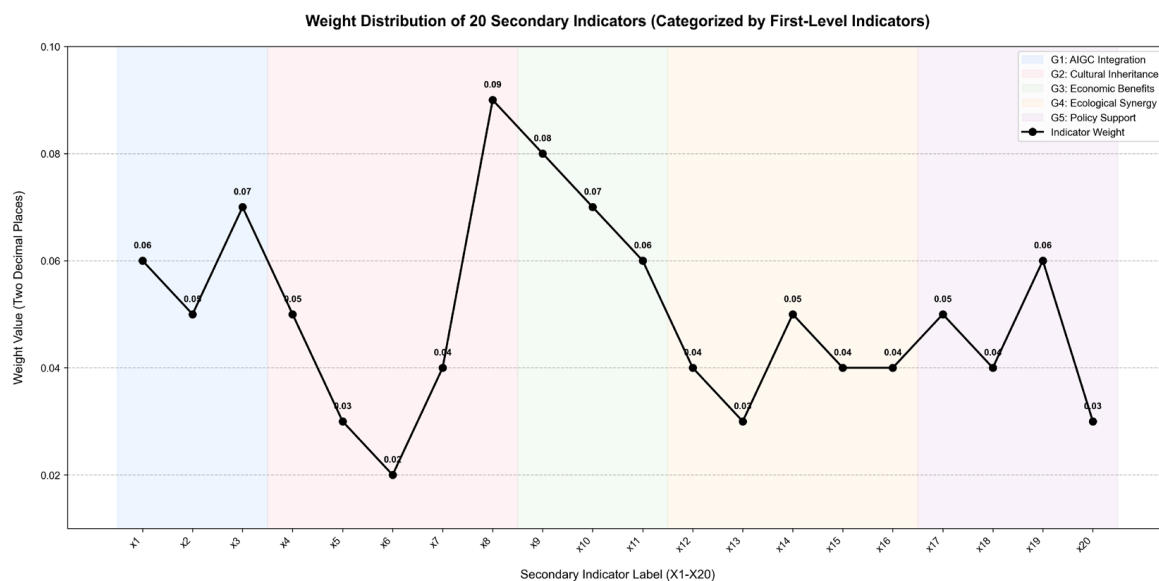
4. Results

4.1. Indicator Weight

Based on the IFAHP method, the judgment matrix is constructed and consistency tested ($CR < 0.1$ under all conditions, meeting the consistency test), and the final weights of first-level and key second-level indicators are calculated, as shown in Table 6 and Figure 7.

Table 6. Weights of first-level and second-level indicators.

First-Level Indicator	Weight of First-Level Indicator	Second-Level Indicator	Weight of Second-Level Indicator
AIGC Technology Integration G_1	0.18	x_1 : AIGC Technology Penetration Depth	0.06
		x_2 : Virtual Agglomeration Synergy Efficiency	0.05
		x_3 : Technology Iteration Resilience	0.07
Cultural Inheritance and Protection G_2	0.23	x_4 : Digital Coverage Rate of Traditional Culture	0.05
		x_5 : Public Participation Rate in Inheritance	0.03
		x_6 : Integrity of Digital Cultural Resources	0.02
		x_7 : Integration Rate of Traditional Elements	0.04
		x_8 : Intergenerational Continuity of Cultural Inheritance	0.09
Economic Benefits of Digital Cultural Industry G_3	0.21	x_9 : Creative Industry GDP Contribution Rate	0.08
		x_{10} : Economic Output of Value Co-Creation	0.07
		x_{11} : Employment-Driving Effect	0.06
Ecological Synergy and Social Inclusiveness G_4	0.20	x_{12} : Multi-Subject Collaboration Frequency	0.04
		x_{13} : Digital Skill Training Coverage Rate	0.03
		x_{14} : Green Energy Utilization Rate	0.05
		x_{15} : Public Satisfaction Degree	0.04
		x_{16} : Low-Carbon Operation Level	0.04
Governance and Policy Support G_5	0.17	x_{17} : Intellectual Property Protection Degree	0.05
		x_{18} : Policy Support Intensity	0.04
		x_{19} : Digital Service Level	0.06
		x_{20} : Cross-Subject Collaborative Governance Efficiency	0.03

**Figure 7.** All indicators' weights.

After the weighted normalized matrix and ideal solutions are obtained, the Euclidean distances and closeness degrees of the five sample cities are calculated, and the final ranking is obtained (see Table 7 and Figure 8).

Table 7. TOPSIS ranking results of sample cities.

Cities	Distance to Positive Ideal Solution D_i^+	Distance to Negative Ideal Solution D_i^-	Closeness Degree $\bar{C}_i$	Ranking of Cities
London	0.0877	0.172	0.6623	1
Shanghai	0.1345	0.1506	0.5282	2
Los Angeles	0.1555	0.1521	0.4945	3
Tokyo	0.162	0.11	0.4043	4
Berlin	0.2189	0.0714	0.2459	5

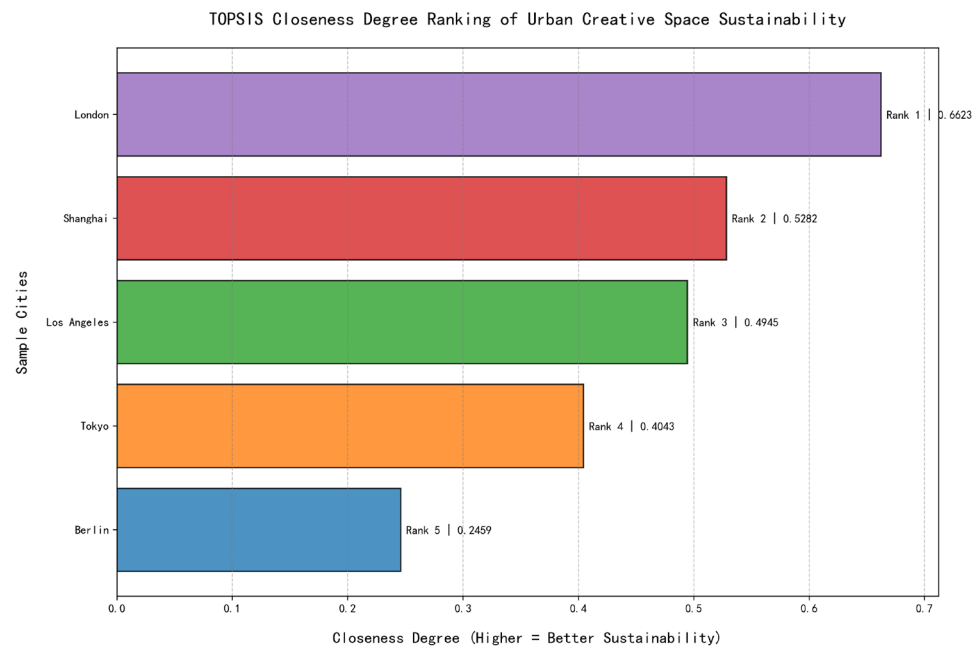


Figure 8. Closeness degree among 5 cities.

4.2. Urban Sustainability Ranking

Based on the TOPSIS method, the sustainable development levels of the five sample cities were comprehensively evaluated and ranked using the indicator weights derived from the IFAHP.

4.2.1. Dimensional Performance of Sample Cities

The five cities exhibit distinct performance patterns across the five primary dimensions. In AIGC technology integration (G_1), London achieves the highest average score (87.3), followed by Los Angeles (84.3), Shanghai and Tokyo (both 73.0), and Berlin (68.0). London's leading position is driven by its outstanding performance in AIGC technology penetration depth ($x_1 = 88$) and virtual agglomeration synergy efficiency ($x_2 = 90$).

In cultural inheritance and protection (G_2), Shanghai ranks first, with an average score of 80.2, benefiting from its high digital coverage rate of traditional culture ($x_4 = 85$) and strong intergenerational continuity ($x_8 = 88$). London follows with 74.0, while Los Angeles (64.4), Tokyo (60.8), and Berlin (53.4) lag behind.

For Economic Benefits of Digital Cultural Industry (G_3), Los Angeles demonstrates clear dominance, with an average score of 87.3, driven by its exceptional Creative Industry GDP Contribution Rate ($x_9 = 92$) and economic output of value co-creation ($x_{10} = 90$). London (81.0) and Shanghai (72.0) occupy the middle positions, while Tokyo (65.0) and Berlin (61.0) score lower.

In ecological synergy and social inclusiveness (G_4), Tokyo leads with an average score of 84.8, supported by its highest green energy utilization rate ($x_{14} = 90$) and low-carbon operation level ($x_{16} = 92$). Berlin follows with 78.4, then London (72.8), Shanghai (70.6), and Los Angeles (66.6).

For governance and policy support (G_5), Shanghai achieves the highest average score (80.2), reflecting its strong policy support intensity ($x_{18} = 85$) and Digital Public Service Level ($x_{19} = 82$). London (78.6) and Los Angeles (72.4) follow, with Tokyo (70.8) and Berlin (69.2) ranking lower.

4.2.2. Comprehensive Ranking Results

The Euclidean distances to the positive ideal solution (D^+) and negative ideal solution (D^-), along with the relative closeness degrees (C), were calculated for each city. London exhibits the shortest distance to the positive ideal solution ($D^+ = 0.0877$) and the longest distance to the negative ideal solution ($D^- = 0.1720$), yielding the highest closeness degree ($C = 0.6623$) and ranking first.

Shanghai shows a D^+ of 0.1345 and D^- of 0.1506, with a closeness degree of 0.5282, ranking second. Los Angeles has a D^+ of 0.1555 and D^- of 0.1521, resulting in a closeness degree of 0.4945 and third place. Tokyo presents a D^+ of 0.1620 and D^- of 0.1100, with a closeness degree of 0.4043, ranking fourth. Berlin exhibits the longest distance to the positive ideal solution ($D^+ = 0.2189$) and the shortest distance to the negative ideal solution ($D^- = 0.0714$), achieving the lowest closeness degree ($C = 0.2459$) and ranking fifth.

The comprehensive ranking results indicate that London demonstrates the highest level of sustainable development in urban creative spaces among the five sample cities, followed by Shanghai, Los Angeles, Tokyo, and Berlin. This ordering reflects the aggregated performance differences across the five dimensions of technology integration, cultural preservation, economic benefits, ecological and social inclusiveness, and governance support.

5. Discussion

The results of this study are of analytical value. From these results, one can understand the role of policies, economic measures, and other factors in the development of urban creative spaces. This not only reflects the core logic of the sustainable development of urban creative spaces, but also confirms the research contribution proposed in Section 1: using cultural inheritance and technological innovation as the core driving dimensions balances the supporting roles of economy, ecology, and governance.

5.1. Interpretation of Indicator Weights

The weight distribution of 20 secondary indicators, determined by the IFAHP, reflects the relative importance of each dimension in evaluating the sustainability of urban creative spaces. From the quantitative results, the weight allocation presents clear hierarchical characteristics and practical rationality.

In terms of first-level indicators, the weight of cultural inheritance and protection (G_2) is the highest, accounting for 23% of the total weight. Among its secondary indicators, Intergenerational Continuity of Cultural Inheritance (x_8) obtains the highest individual weight, followed by digital coverage rate of traditional culture (x_4) with a weight of 0.05. This indicates that maintaining cultural vitality across generations and promoting the digital preservation of traditional culture are core elements of sustainable development, which aligns with the inherent requirement of urban creative spaces to retain cultural uniqueness.

The second highest weight is AIGC technology integration (G_1). Among its second-level indicators, technology iteration resilience (x_3 , which is 0.07) and AIGC technology penetration depth (x_1 , which is 0.06) are key indicators, reflecting the irreplaceable role of technological innovation in driving the transformation of creative spaces.

The weights of Economic Benefits of Digital Cultural Industry (G_3) and ecological synergy and social inclusiveness (G_4) are relatively balanced, accounting for 17% and 21%, respectively. Among their indicators, Creative Industry GDP Contribution Rate (x_9) and low-carbon operation level (x_{16}) are the core indicators of the two dimensions. Material guarantee and ecological harmony are indispensable for sustainable development.

Governance and policy support (G_5) accounts for 21%; intellectual property protection degree (x_{17}), as the key indicator, emphasizes the institutional guaranteed role of policy and governance.

From the perspective of individual secondary indicators, the weights of x_8 (0.09), x_9 (0.08), and x_3 (0.07) are among the top three, while x_6 (integrity of digital cultural resources, 0.02) and x_5 (public participation rate in inheritance, 0.03) have relatively low weights. This distribution is not only based on the expert evaluation matrix but also consistent with the actual development focus of urban creative spaces, prioritizing core driving forces such as cultural inheritance, technological adaptation, and economic output while not ignoring supporting factors such as resource management and public participation.

Overall, the weight system effectively balances the multi-dimensional needs of sustainable development and provides a scientific basis for the subsequent evaluation.

5.2. Comparative Analysis of Sample City Performance

Based on the TOPSIS method, the sustainability evaluation results of the five sample cities are ranked by closeness degree as follows: London (0.6623), Shanghai (0.5282), Los Angeles (0.4945), Tokyo (0.4043), and Berlin (0.2459). The detailed interpretation of the evaluation conclusions, combined with specific indicator data, the relevant literature, and practical policies, is as follows.

(a) London: Leading in Technological Integration and Cultural Adaptation

London ranks first with the highest closeness degree (0.6623), showing outstanding performance in multiple dimensions. In AIGC technology integration (G_1), its indicators x_1 (AIGC technology penetration depth, 88) and x_2 (virtual agglomeration synergy efficiency, 90) are the highest among all cities. This is closely related to London's active promotion of AIGC application in the creative industry through the "Creative Industries Sector Deal 2023" and the "AI for Creative Industries Fund", which provides funding for AI co-creation tool development and enterprise adoption. Creative professionals in London widely use AI tools for content generation and immersive experience design, and the government's "AI Intellectual Property Framework" provides clear copyright protection guidelines, creating a favorable environment for technological integration. In cultural inheritance and protection (G_2), x_8 (intergenerational continuity, 73) performs well, benefiting from projects which link young designers with cultural heritage institutions to promote the inheritance of traditional crafts. However, x_5 (Public Participation Intensity, 72) is relatively low, indicating that despite initiatives like the "Digital Cultural Access Scheme", there is still room for improvement in encouraging widespread public participation in cultural digitalization.

The creative industries in London contributed 14.8% to the GDP in 2023. Among them, AIGC-related creative industries grew at a rate of 27%. This economic reality is highly consistent with the results of this study, where the London AIGC technological integration dimension scored first and the economic benefit dimension ranked second.

(b) Los Angeles: Dominant in Economic Benefits, with Ecological Optimization Needs

Los Angeles ranks third with a closeness degree of 0.4945, and its core advantage lies in Economic Benefits of Digital Cultural Industry (G_3). The indicators x_9 (Creative Industry GDP Contribution Rate, 92) and x_{10} (economic output of value co-creation, 90) are far ahead of other cities, which is consistent with the characteristics of Los Angeles' digital creative industry as a key economic pillar. The city's value co-creation model, supported by policies such as California's "Film and Television Tax Incentive Program" (providing up to 30% tax rebates for film and television projects) and the "Hollywood Innovation Cluster Plan", has formed a mature industrial chain spillover effect.

However, Los Angeles' performance in ecological synergy and social inclusiveness (G_4) is insufficient: x_{14} (green energy utilization rate, 55) and x_{16} (low-carbon operation level, 58) are the lowest among the five cities. This reflects the ecological pressure brought by the rapid development of its creative industry, as the high energy consumption of film production studios and cultural entertainment facilities has not been effectively regulated by policies in the creative sector.

The Hollywood-based Los Angeles film and television creative industry created over 200,000 jobs in 2023, contributing 12.3% to the GDP. This result is consistent with the finding in this study that Los Angeles ranked first in terms of economic performance. However, the industry's green energy utilization rate of only 32% explains the low score in the environmental dimension.

(c) Shanghai: Strong Policy Support, to Enhance Cultural and Technological Integration

Shanghai's closeness degree of 0.5282 ranks second, with remarkable advantages in governance and policy support (G_5). x_{18} (policy support intensity, 85) and x_{19} (Digital Public Service Level, 82) are among the top, attributed to China's "14th Five-Year Plan for Digital Cultural Industry" and Shanghai's "Digital Creative Industry Cluster Development Plan", which provide special funds, land support, and one-stop digital government services for creative enterprises. In cultural inheritance and protection (G_2), x_4 (digital coverage rate of traditional culture, 85) is the highest, thanks to projects such as the "Digital Protection Project of Shanghai Intangible Cultural Heritage", which has digitized over 100 intangible cultural heritage items.

Shanghai's x_1 (AIGC technology penetration depth, 75) and x_3 (technology iteration resilience, 68) are relatively low compared with London and Los Angeles. This is partly due to the "Interim Measures for the Administration of Generative Artificial Intelligence Services", which imposes stricter content review requirements on AIGC applications, limiting the speed of technological iteration in creative scenarios, and partly due to the lack of deep cooperation between local creative enterprises and AI technology providers.

In 2024, the revenue of digital creative industry enterprises in Shanghai exceeded 500 billion yuan. The digitalization projects of intangible cultural heritage led to an 18% increase in the income of the cultural tourism industry, confirming the high score of the cultural inheritance and policy support dimensions in this research.

(d) Tokyo: Outstanding Ecological Synergy, Weak in Cultural Innovation Integration

Tokyo ranks fourth, with a closeness degree of 0.4043, and its highlight is ecological synergy and social inclusiveness (G_4). x_{14} (green energy utilization rate, 90) and x_{16} (low-carbon operation level, 92) are the highest among all cities, closely related to Tokyo's "Urban Low-Carbon Renewal Strategy". The strategy requires creative parks to use at least 50% renewable energy, and certified low-carbon creative spaces can receive rent subsidies. The city's promotion of renewable energy and circular economy policies has effectively supported the ecological sustainability of creative spaces. However, Tokyo's performance in cultural inheritance and protection (G_2) and AIGC technology integration (G_1) is average: x_7 (integration rate of traditional elements, 63) and x_2 (virtual agglomeration synergy efficiency, 69) are relatively low. This reflects the challenge of integrating traditional cultural elements with modern technological innovation, as Tokyo's "Cultural Heritage Protection Act" emphasizes the preservation of traditional forms but lacks policies encouraging the innovative application of cultural elements in AIGC scenarios.

Tokyo scored relatively low in terms of cultural heritage protection and digitization. The core reason for this lies in two major challenges. Technically, the digitization of Tokyo's cultural heritage is still at the "archival" stage and has not applied AIGC (artificial intelligence, Internet of Things, and Cloud Computing) technology to the innovative

transformation of cultural elements. Culturally, the “Cultural Heritage Protection Law” overly emphasizes the “authenticity protection” of traditional forms and lacks policy guidance that encourages the integration of cultural heritage with modern creativity. This is the main reason why its cultural inheritance dimension score is lower than that of Shanghai and London.

(e) Berlin: Balanced Development, Lack of Core Advantages

Berlin’s closeness degree of 0.2459 ranks fifth. The city shows relatively balanced performance in all five dimensions but lacks core competitive advantages. In ecological synergy and social inclusiveness (G_4), x_{13} (Digital Skill Popularization Level, 85) is outstanding, reflecting Berlin’s efforts in reducing the digital divide through the “Digital Inclusion Program for Creative Industries”, which provides free digital skill training for low-income groups and independent creators. However, Berlin’s indicators in AIGC technology integration (G_1) and Economic Benefits of Digital Cultural Industry (G_3) are low ($x_1 = 68$, $x_9 = 60$). This is related to Germany’s relatively conservative attitude towards AIGC technology (reflected in the local AI Ethics Guidelines) and the lack of targeted industrial support policies—unlike other cities, Berlin does not have special tax incentives or funding support for digital creative enterprises, leading to slow industry development compared with other international metropolises.

5.3. Common Challenges and Optimization Pathways

By analyzing the indicator scores and evaluation results of the five sample cities, several common problems affecting the sustainable development of urban creative spaces are identified, which need to be addressed in future practice.

i. Insufficient Depth of Public Participation in Cultural Inheritance

Public participation rate in inheritance (x_5) shows relatively low scores across all cities: Los Angeles and Berlin are at a low level, while even London and Shanghai have room for improvement, which indicates that the current participation of the public is not sufficient [10]. The lack of in-depth public participation not only weakens the social identity of cultural heritage but also affects the sustainability of cultural inheritance. There is an urgent need for innovative participation mechanisms based on human-centered design.

Innovative public participation mechanisms play a core role in enhancing social sustainability. On one hand, digital cultural participation scenarios based on AIGC (such as online traditional craft creation) can lower the threshold for public participation, enhance the social recognition of cultural heritage, and strengthen the social cohesion of urban creative spaces. On the other hand, the diverse participation of the public can provide more ideas for cultural heritage innovation and promote the sustainability of cultural inheritance.

ii. Digital Skill Divide Restricts Social Inclusiveness

Except for Berlin and Tokyo, the digital skill training coverage rate (x_{13}) of other cities, Shanghai, Los Angeles, and London, is relatively low. This reflects that digital skill training for low-income groups and groups with weak digital skills is not in place, leading to a digital divide in the use of creative space resources. With the deepening of AIGC technology integration, the demand for digital skills will further increase, and the solution to this problem has become an important prerequisite for realizing social sustainability.

Digital skills training contributes to the sustainability of society in two ways. Firstly, digital skills training for disadvantaged groups can reduce the digital divide, allowing more groups to share the resources of urban creative spaces. Secondly, the popularization of digital skills can cultivate more creative participants, strengthen the social foundation, and achieve sustainable development at the societal level. For example, the digital skills training program in Berlin has verified that when the coverage rate of digital skills training

increases, the public satisfaction of creative spaces will also rise, and social inclusiveness will be significantly enhanced.

iii. Imperfect Integrity of Digital Cultural Resources

The integrity of digital cultural resources (x_6) has an average score across the five cities, indicating that there are problems such as non-standardized archiving and poor cross-platform compatibility of digital cultural resources. The imperfect integrity of digital resources not only affects the long-term reuse of cultural heritage but also restricts the in-depth integration of traditional culture and creative industries. In the context of the rapid development of digitalization, improving the standardization and integrity of digital cultural resources has become a common task for urban creative spaces.

i. Policy differences reflect different directions of development.

London and Shanghai, as exemplary cities for the sustainable development of creative spaces, exhibit significant regional differences in their policy support systems. London relies mainly on market-driven policies, providing precise technical research and development funds through initiatives such as the “AI for Creative Industries Fund”, and focusing on the development of AIGC technologies and intellectual property protection. Shanghai, on the other hand, centers on government-guided policies, relying on the “Shanghai Digital Creative Industry Cluster Development Plan” to offer comprehensive support in terms of land, funds, and taxes, concentrating on the digitization of cultural heritage and the construction of industrial clusters. From the perspective of policy effectiveness, London’s market-driven policies have facilitated the rapid implementation of AIGC technologies (corresponding to the high score of the AIGC dimension in this study), while Shanghai’s government-guided policies have achieved a deep integration of cultural heritage preservation and policy support (corresponding to the high scores of the cultural heritage preservation and policy support dimensions in this study). The policy practices of these two cities provide two typical policy paradigms for the sustainable development of creative spaces in global cities.

ii. The technical and cultural challenges of heritage digitization

All sample cities encounter common challenges during the process of heritage digitization. Technically, some cities’ digitization technologies only achieve “online migration” of cultural heritage, lacking the deep integration of technologies such as AIGC, resulting in a low reuse rate of digital resources. Culturally, some cities are caught in the binary opposition of “authenticity protection” and “innovative application”, failing to establish a mechanism for balancing the inheritance of traditional culture and the development of modern creativity. This challenge is particularly prominent in cities like Tokyo.

These common problems are not isolated but interrelated. For example, the digital skill divide will further restrict public participation, and the imperfect integrity of digital resources will affect the effect of cultural inheritance. Solving these problems requires the joint efforts of the government, enterprises, and the public, and provides a direction for subsequent optimization of the sustainable development path of urban creative spaces.

5.4. Model Robustness and Methodological Implications

To address reviewers’ concerns regarding methodological validity, specifically the stability of weight-based rankings and the reliability of claims about global applicability, this study conducts robustness verification through weight perturbation analysis. The goal is to test whether the relative ranking of sample cities remains consistent when indicator weights fluctuate within a reasonable range, thereby validating the scientific rigor and practical reliability of the constructed evaluation framework.

Specifically, the sensitivity analysis focuses on perturbing the weights of the five first-level indicators, a design aligned with the inherent uncertainty of expert judgments in complex multi-attribute decision-making. For each first-level indicator, we applied $\pm 10\%$ and $\pm 20\%$ perturbation amplitudes. The weights of secondary indicators under the same first-level indicator were scaled proportionally to ensure the total weight sum remained one. Spearman's rank correlation coefficient (ρ) was adopted as the key evaluation criterion, where ρ ranges from -1 to 1 . $\rho \geq 0.8$ indicates high consistency between perturbed and original rankings, $\rho = 1$ denotes identical rankings, and lower ρ values reflect greater ranking volatility.

The detailed results of the sensitivity analysis are presented in Table 8, with the visualization of Spearman's rank correlation coefficients shown in Figure 9. As illustrated, the robustness verification yields two prominent characteristics. First, the evaluation framework exhibits high stability under most perturbation scenarios: 14 out of 20 scenarios (70%) yield a Spearman's ρ of 1.0, meaning the urban ranking remains identical to the original IFAHP-based order (London > Shanghai > Los Angeles > Tokyo > Berlin). Notably, perturbations to the weights of AIGC technology integration, ecological synergy and social inclusiveness, and governance and policy support, regardless of whether the amplitude is $\pm 10\%$ or $\pm 20\%$, fail to alter the ranking. This indicates strong resistance to weight fluctuations in these dimensions, confirming that the evaluation results are not overly sensitive to reasonable variations in expert judgments of indicator importance.

Table 8. Urban ranking and Spearman's correlation coefficient under weight perturbation.

First-Level Indicator	Perturbation	Shanghai Rank	London Rank	Los Angeles Rank	Tokyo Rank	Berlin Rank	Spearman's ρ
AIGC Technology Integration	−20%	2	1	3	4	5	1.0
	−10%	2	1	3	4	5	1.0
	+10%	2	1	3	4	5	1.0
	+20%	2	1	3	4	5	1.0
Cultural Inheritance and Protection	−20%	2	3	1	4	5	0.6
	−10%	2	1	3	4	5	1.0
	+10%	2	1	3	4	5	1.0
	+20%	2	1	3	4	5	1.0
Economic Benefits of Digital Cultural Industry	−20%	2	1	3	4	5	1.0
	−10%	2	1	3	4	5	1.0
	+10%	2	1	3	4	5	1.0
	+20%	2	3	1	4	5	0.6
Ecological Synergy and Social Inclusiveness	−20%	2	1	3	4	5	1.0
	−10%	2	1	3	4	5	1.0
	+10%	2	1	3	4	5	1.0
	+20%	2	1	3	4	5	1.0
Governance and Policy Support	−20%	2	1	3	4	5	1.0
	−10%	2	1	3	4	5	1.0
	+10%	2	1	3	4	5	1.0
	+20%	2	1	3	4	5	1.0

Furthermore, moderate ranking fluctuations only occur in two extreme perturbation scenarios, both resulting in a Spearman's ρ of 0.6. When the weight of cultural inheritance and protection is reduced by 20% or the weight of Economic Benefits of Digital Cultural Industry is increased by 20%, Los Angeles rises to the top rank while London drops to third. Even in these extreme cases, however, the core ranking of three cities remains unchanged (Shanghai at second, Tokyo at fourth, Berlin at fifth), and the correlation coefficient of 0.6 indicates no fundamental reversal of the overall relative order, only partial adjustment of the top three positions. This fluctuation is logically consistent with the cities' inherent characteristics: reducing the weight of cultural inheritance weakens London's advantage in cultural adaptation, while increasing the weight of economic benefits amplifies Los Angeles' leading performance in the digital creative industry's GDP contribution ($x_9 = 92$), leading to the observed partial ranking shift.

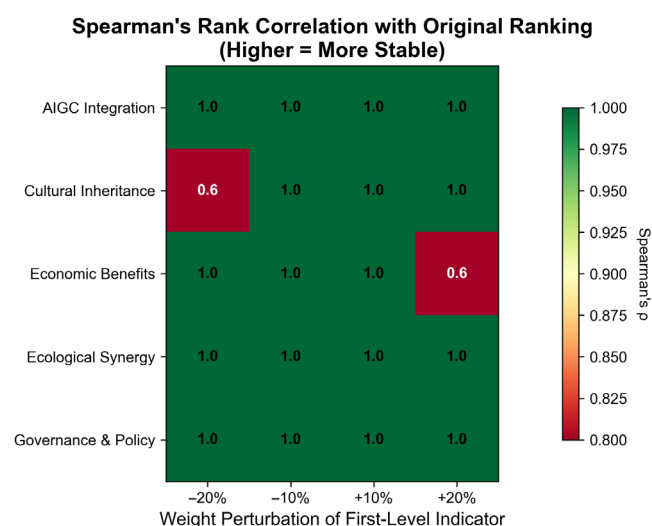


Figure 9. Spearman's rank correlation coefficient between perturbed and original rankings.

The sensitivity analysis confirms that the evaluation framework possesses high robustness. The core ranking of the sample cities remains stable under reasonable weight perturbations, and even extreme weight adjustments only induce partial order shifts rather than fundamental reversals. This stability directly addresses the reviewers' methodological concerns, demonstrating that the ranking results are not contingent on arbitrary weight assignments but reflect substantive differences in sustainable development levels across cities, thus supporting the framework's claim of global applicability with empirical evidence.

Furthermore, the robustness of the results underscores the methodological value of the IFAHP. Unlike the traditional AHP (which simplifies expert judgments into crisp pairwise comparison matrices) or the FAHP (which only considers membership degrees of fuzzy sets), the IFAHP captures the hesitation and ambiguity inherent in expert evaluations through intuitionistic fuzzy numbers, generating more scientifically rigorous weights. As validated by the sensitivity analysis, these weights exhibit strong anti-interference capabilities: even with $\pm 20\%$ perturbations, the core evaluation conclusions remain intact. By refining the handling of expert uncertainty, IFAHP mitigates the risk of biased rankings caused by oversimplified judgment processing, representing a meaningful methodological advancement in decision-making for urban sustainability evaluation.

In summary, the sensitivity analysis validates the stability and reliability of the evaluation results, while the integration of the IFAHP enhances the framework's methodological rigor. These findings provide a solid foundation for subsequent discussions of urban development characteristics and optimization directions.

6. Conclusions

In this study, we aim to address the gap in the systematic assessment of urban creative space sustainability amid global digital transformation and cultural diversity. Centering on the core demand for scientific, practical, and globally referenced assessment tools, it constructs a multi-dimensional evaluation framework, empirically validated through international case studies, to form a complete research chain spanning from theoretical framework to practical application. The research process employs a rigorous combination of qualitative and quantitative methodologies.

First, through a systematic literature review and expert interviews, an evaluation index system is developed, encompassing five primary dimensions, (1) AIGC technology integration, (2) cultural inheritance and protection, (3) Economic Benefits of Digital Cultural Industry, (4) ecological synergy and social inclusiveness, and (5) governance and policy

support, and 20 secondary indicators. Our system fully integrates the dual attributes of technological innovation and cultural inheritance inherent to urban creative spaces, alongside the requirements for balanced economic, ecological, and social development. Second, the Improved Fuzzy Analytic Hierarchy Process (IFAHP) is adopted to determine indicator weights. Compared to the traditional Fuzzy Analytic Hierarchy Process, the IFAHP optimizes the construction of the fuzzy judgment matrix and the consistency test process, which effectively mitigates subjectivity stemming from experts' subjective evaluations while retaining the advantage of handling uncertain information. Weight calculation results reflect the relative importance of each dimension and indicator within the sustainability system, laying a quantitative foundation for subsequent assessments.

6.1. Core Contributions of the Research

In summarizing this study, our innovative contributions are demonstrated across evaluation methodology innovation, theoretical advancement, and case application:

i. Mitigating Subjectivity in Multi-Criteria Decision-Making

This study introduces the IFAHP into the weight calculation process for the evaluation model, largely addressing the FAHP's limitations in handling subjective judgment biases. By refining the fuzzy number conversion rules and enhancing the rationality of consistency tests, the IFAHP reduces the impact of individual experts' preferences on results, making weight allocation more objective and reliable. This methodological optimization not only strengthens the scientific rigor of this study but also provides a viable reference for similar multi-criteria decision-making issues in urban space evaluation.

ii. Constructing a Globally Applicable Evaluation Model

The proposed evaluation index system overcomes the limitations of single-dimensional or region-specific frameworks in existing research. It integrates cutting-edge elements (AIGC-creativity technology integration and digital cultural industry development) with traditional dimensions (cultural inheritance and ecological sustainability). Supported by the extensive literature and academic findings, this forms a comprehensive, systematic evaluation framework. With empirical analysis of five cities with distinct cultural backgrounds, economic bases, and development models, the model demonstrates adaptability across diverse urban contexts, filling the gap in globally applicable tools for assessing urban creative space sustainability.

iii. Integrating Global Cases with Expert Panel Scoring

This study collects primary data from five world-renowned creative cities, covering their status in technological application, cultural protection, economic benefits, ecological construction, and policy support. The expert scoring process involves interdisciplinary professionals (from urban planning, cultural industries, and digital technology), who provide in-depth assessments and constructive insights based on practical experience. This integration of multi-city empirical data and expert wisdom not only validates the model's effectiveness but also provides a comparative benchmark for urban managers to formulate sustainable development strategies.

6.2. Limitations and Future Research Directions

While this study contributes to both theory and practice, there remains room for improvement.

- (a) The sample size of evaluated cities can be expanded. This study selects five representative cities, but urban creative space development varies across regions (like emerging creative cities in Asia, Africa, and Latin America). Notably, the current sample, though typologically representative, limits the direct generalization of conclusions to

global contexts, and future research should include more cities from underrepresented regions to enhance the framework's cross-cultural adaptability.

- (b) A dynamic assessment dimension can be added. We use a static, cross-sectional data-based approach, which struggles to capture temporal changes in urban creative space sustainability. Methodologically, the reliance on expert judgment for certain indicators introduces inevitable subjectivity, and future studies could combine longitudinal tracking with mixed methods (e.g., ethnography) to complement cross-sectional data and validate causal relationships.
- (c) The integration of emerging technical tools can be strengthened. With advances in big data and artificial intelligence, future research can use data mining to collect real-time data (e.g., creative space user behavior, energy consumption) to replace part of expert scoring, improving the timeliness and accuracy of evaluation results.

In conclusion, in the study, we construct a scientific, applicable sustainability evaluation framework for urban creative spaces through methodological optimization, theoretical construction, and empirical validation. The findings not only enrich the theoretical system of urban creative space research but also provide practical guidance for global urban creative space sustainability. As research advances and application scenarios expand, this model will continue to be refined, better serving the high-quality development of urban creative industries in the digital era.

Supplementary Materials: The sources of Figures 3 and 6 in the article, and have listed the links below. Figure 3: News report from the Chinese Culture Conservation Project of the College of Fashion and Design at Donghua University. <https://yurenhao1.sizhengwang.cn/a/dhdx/240918/1921149.shtml>. The links for Figure 6a–e are as follows: (a) Shanghai: <https://bbs.co188.com/thread-10209909-1-1.html>. (b) London: <https://hubblehq.com/blog/the-best-creative-office-spaces-in-london>; (c) Los Angeles: <https://www.graymag.com/post/officeuntitled-ava-arts-district>; (d) Japan: <https://asia.nikkei.com/life-arts/life/tokyo-s-new-urban-spaces-want-you-to-spend-time-not-just-money>; (e) Berlin: <https://culture.ec.europa.eu/cultural-and-creative-sectors/architecture/living-spaces/catalogue/berlin>. (links are accessed on 17 March 2026).

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Abbreviations

The following abbreviations are used in this manuscript:

TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
IFAHP	Intuitionistic Fuzzy Analytic Hierarchy Process
AIGC	Artificial Intelligence-Generated Content
CCI	Cultural Creative Industry

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